

1 Missing Values

What you found:

Some records had missing information, especially student names, grades, return dates, and some book information. Some missing values also came from the 26 HTML records because they did not contain all database columns.

Where it was:

The problem appeared in columns such as `first_name`, `last_name`, `grade`, `return_date`, `publication_year`, and other columns.

How big it was:

There were missing values in several columns. For example, `first_name` had 26 missing values, `grade` had 62, `return_date` had 91, and `publication_year` had 60.

What you did, and why:

We filled missing text values with "Unknown" to keep the records without deleting information. We kept missing `return_date` values because an empty return date can mean that the book has not been returned yet.

2 Duplicate Records

What you found:

Some rows were completely identical, meaning the same record appeared more than once.

Where it was:

The problem affected the complete rows across the dataset.

How big it was:

There were **8 duplicate records**.

What you did, and why:

We used `drop_duplicates()` to remove the 8 identical records. This prevents the same checkout from being counted more than once and keeps the analysis accurate.

3 Inconsistent Text

What you found:

Neighborhood names were written in different ways, including extra spaces and different capitalization. For example, the same neighborhood could appear as `maadi`, `MAADI`, or `Maadi`.

Where it was:

The problem appeared in the `neighborhood` column.

How big it was:

There were **10 inconsistent values**, which were reduced to **6 clean neighborhood names**.

What you did, and why:

We used `.str.strip()` to remove extra spaces and `.str.title()` to standardize capitalization. This makes the neighborhood values consistent and improves the accuracy of grouping and counting.

4 Orphan Records

What you found:

Some checkout records could not be matched to a registered member. These are called **orphan records** because the checkout exists, but there is no matching member information.

Where it was:

The problem appeared when checking the `member_id` column after combining the checkout and member data.

How big it was:

The number of orphan records was identified by checking the missing `member_id` values. **The exact number should be taken from the result of the Python code.**

What you did, and why:

We kept the orphan records instead of deleting them and used `"Unknown"` for the missing member information. This preserves the checkout records and prevents losing useful library data.